FairCKD: Fairness Audit Report

Investigating Bias and Fairness in Machine Learning Models for Automated CKD Diagnosis
M.Tech Dissertation Phase-II · Jimil Jingar · PIET, Parul University

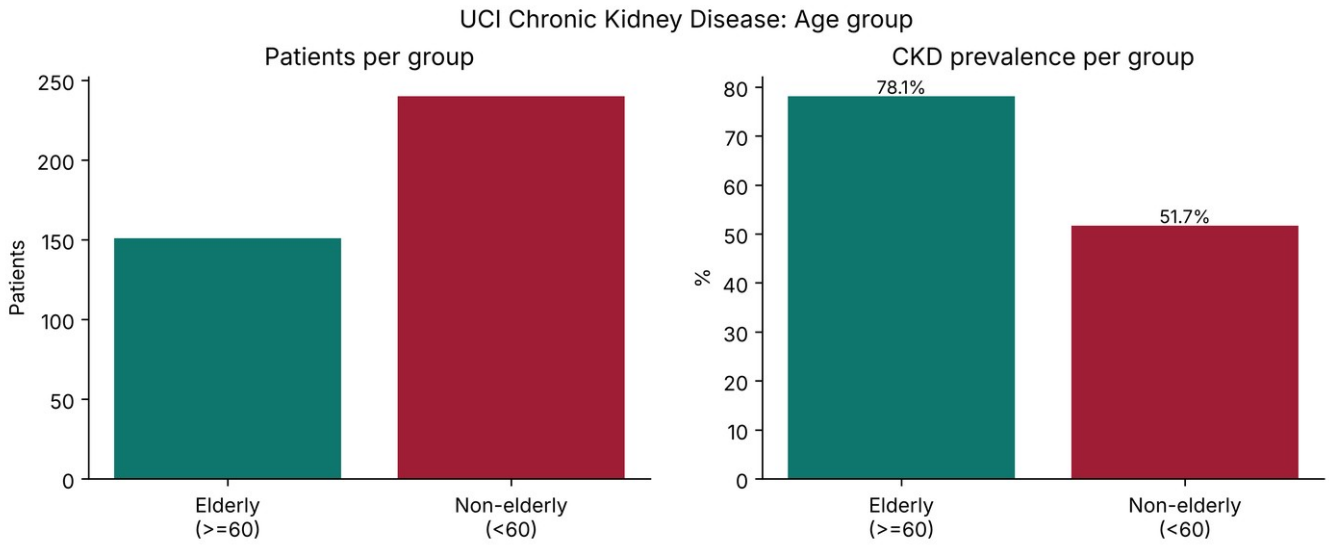
	Value
Dataset	UCI Chronic Kidney Disease
Feature set	Primary-care screening (10 features, no kidney labs)
Sensitive attribute	Age group
Patients	391
Features	10
Groups	Elderly (>=60), Non-elderly (<60)
Decision threshold	0.500
CV protocol	stratified 5-fold, 10 repeats (baseline), 5 repeats (mitigation)

Key findings

• Best baseline model: **Random Forest** (accuracy 0.940, balanced accuracy 0.949, AUC 0.978).
• Bias detection: in **7 of 7** models, CKD is detected least often in the **Non-elderly (<60)** group (Equal Opportunity gap 0.035–0.138).
• Even a perfect classifier has DP diff = 0.265 and DI = 0.661 here, because CKD prevalence differs between groups; error-rate metrics (Equal Opportunity, Equalized Odds) are the clinically meaningful fairness criteria.
• Logistic Regression: best mitigation (balanced accuracy loss ≤ 3 points) = **Threshold Opt.**, Equalized Odds gap 0.081 → 0.038, balanced accuracy 0.929 → 0.911.
• Random Forest: best mitigation (balanced accuracy loss ≤ 3 points) = **Threshold Opt.**, Equalized Odds gap 0.035 → 0.029, balanced accuracy 0.948 → 0.937.
• XGBoost: best mitigation (balanced accuracy loss ≤ 3 points) = **Threshold Opt.**, Equalized Odds gap 0.089 → 0.022, balanced accuracy 0.904 → 0.895.
• **FairCKD-Select** (proposed, nested CV): Equalized Odds gap 0.084 → **0.043** compared with accuracy-driven model selection, balanced accuracy 0.928 → 0.908.
• Statistical significance: 0 of 15 mitigation results reduce the Equalized Odds gap significantly (paired bootstrap, $p < 0.05$).

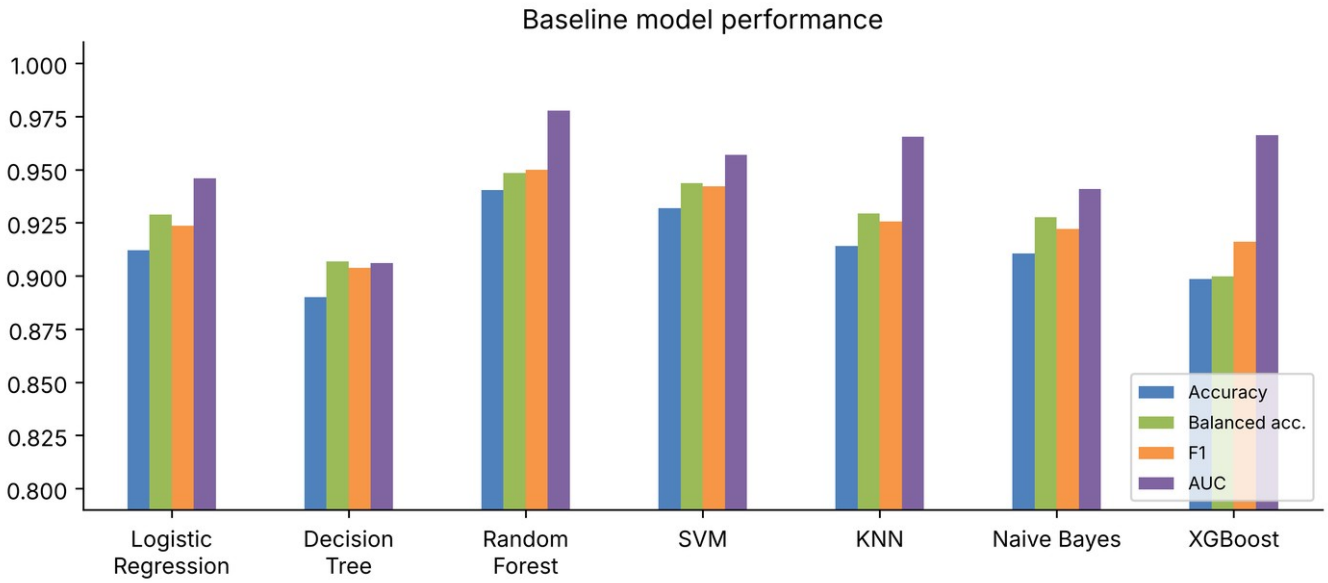
1. Dataset and groups

Group	Patients	Share (%)	CKD cases	CKD prevalence (%)	Missing values / patient
Elderly (>=60)	151	38.6	118	78.1	0.280
Non-elderly (<60)	240	61.4	124	51.7	0.280
All	391	100	242	61.9	0.280


UCI Chronic Kidney Disease: Age group
Patients per group
Patients
250
200
150
100
50
0
Elderly
(>=60)
Non-elderly
(<60)
CKD prevalence per group
%
80
70
60
50
40
30
20
10
0
78.1%
51.7%
Elderly
(>=60)
Non-elderly
(<60)

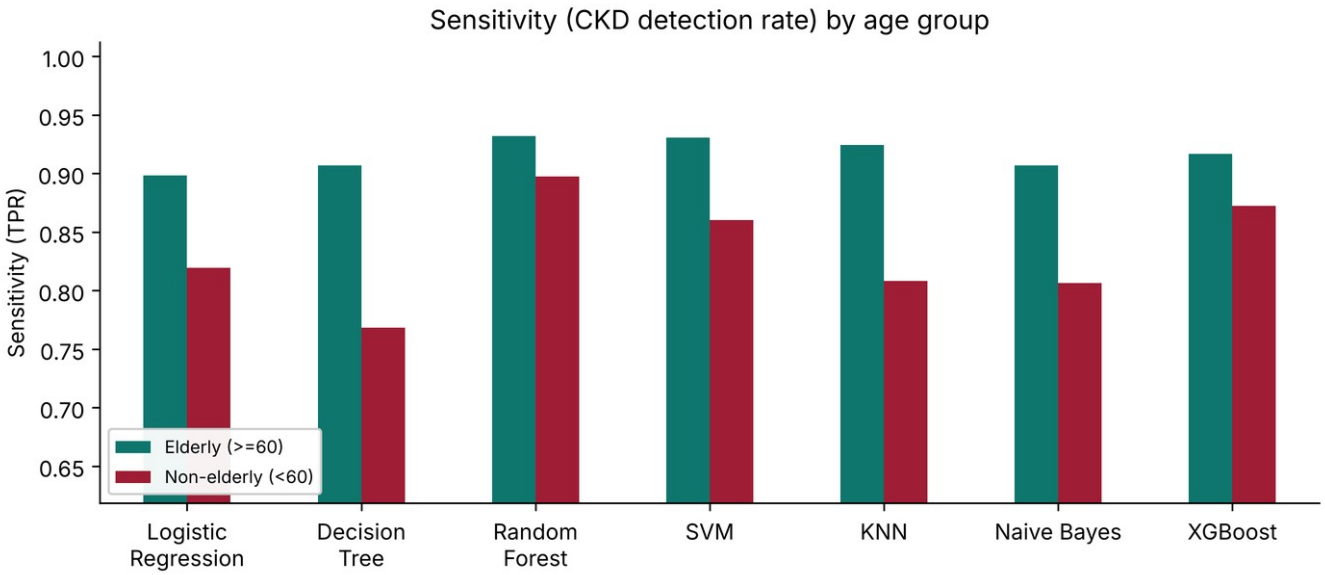
2. Baseline model performance

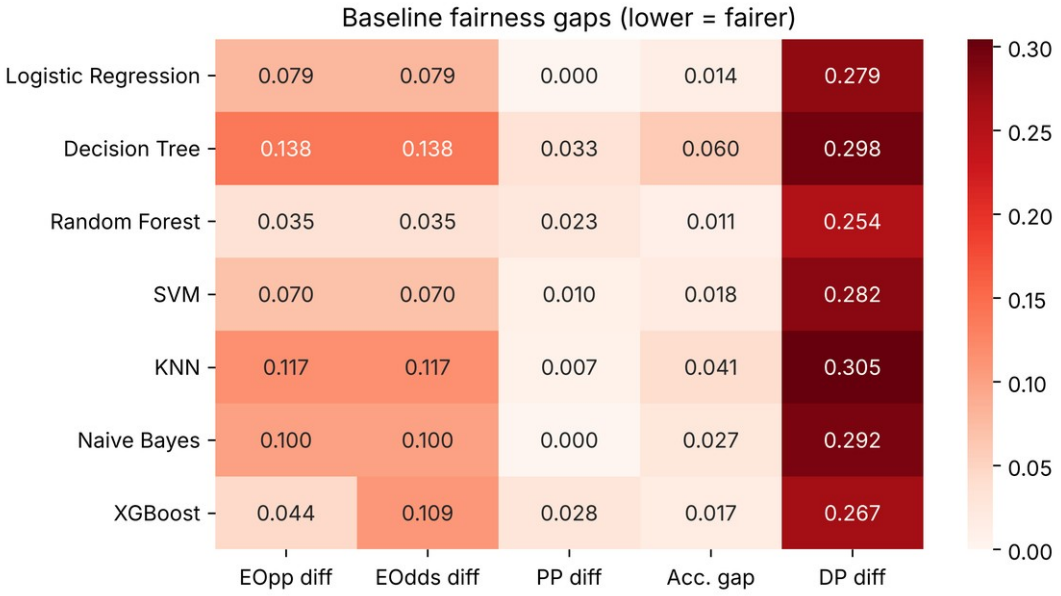
	Accuracy	Precision	Recall	F1	AUC	Balanced acc.
Logistic Regression	0.912 ± 0.001	1.000 ± 0.000	0.858 ± 0.002	0.923 ± 0.001	0.946 ± 0.004	0.929 ± 0.001
Decision Tree	0.890 ± 0.007	0.984 ± 0.004	0.836 ± 0.013	0.904 ± 0.007	0.906 ± 0.006	0.907 ± 0.005
Random Forest	0.940 ± 0.004	0.988 ± 0.004	0.914 ± 0.005	0.950 ± 0.004	0.978 ± 0.003	0.949 ± 0.004
SVM	0.932 ± 0.006	0.995 ± 0.003	0.895 ± 0.007	0.942 ± 0.005	0.957 ± 0.003	0.944 ± 0.005
KNN	0.914 ± 0.004	0.996 ± 0.004	0.865 ± 0.006	0.926 ± 0.004	0.965 ± 0.004	0.929 ± 0.004
Naive Bayes	0.910 ± 0.000	1.000 ± 0.000	0.855 ± 0.000	0.922 ± 0.000	0.941 ± 0.002	0.928 ± 0.000
XGBoost	0.898 ± 0.008	0.939 ± 0.009	0.894 ± 0.008	0.916 ± 0.007	0.966 ± 0.003	0.900 ± 0.009



3. Bias detection and fairness evaluation

	DP diff	DI ratio	EOpp diff	EOdds diff	PP diff	Acc. gap
Logistic Regression	0.279 ± 0.002	0.603 ± 0.003	0.079 ± 0.004	0.079 ± 0.004	0.000 ± 0.000	0.014 ± 0.002
Decision Tree	0.298 ± 0.015	0.580 ± 0.016	0.138 ± 0.022	0.138 ± 0.022	0.033 ± 0.008	0.060 ± 0.016
Random Forest	0.254 ± 0.005	0.651 ± 0.007	0.035 ± 0.009	0.035 ± 0.009	0.023 ± 0.007	0.011 ± 0.007
SVM	0.282 ± 0.007	0.613 ± 0.007	0.070 ± 0.011	0.070 ± 0.011	0.010 ± 0.007	0.018 ± 0.012
KNN	0.305 ± 0.007	0.580 ± 0.006	0.117 ± 0.012	0.117 ± 0.012	0.007 ± 0.005	0.041 ± 0.010
Naive Bayes	0.292 ± 0.000	0.588 ± 0.000	0.100 ± 0.000	0.100 ± 0.000	0.000 ± 0.000	0.027 ± 0.000
XGBoost	0.267 ± 0.020	0.646 ± 0.021	0.044 ± 0.013	0.109 ± 0.047	0.028 ± 0.024	0.017 ± 0.010
Perfect classifier (reference)	0.265	0.661	0.000	0.000	0.000	0.000


Sensitivity (CKD detection rate) by age group
Sensitivity (TPR)
1.00
0.95
0.90
0.85
0.80
0.75
0.70
0.65
Elderly (>=60)
Non-elderly (<60)
Logistic Regression
Decision Tree
Random Forest
SVM
KNN
Naive Bayes
XGBoost


Baseline fairness gaps (lower = fairer)
Logistic Regression 0.079 0.079 0.000 0.014 0.279
Decision Tree 0.138 0.138 0.033 0.060 0.298
Random Forest 0.035 0.035 0.023 0.011 0.254
SVM 0.070 0.070 0.010 0.018 0.282
KNN 0.117 0.117 0.007 0.041 0.305
Naive Bayes 0.100 0.100 0.000 0.027 0.292
XGBoost 0.044 0.109 0.028 0.017 0.267
EOpp diff
EOdds diff
PP diff
Acc. gap
DP diff
0.30
0.25
0.20
0.15
0.10
0.05
0.00

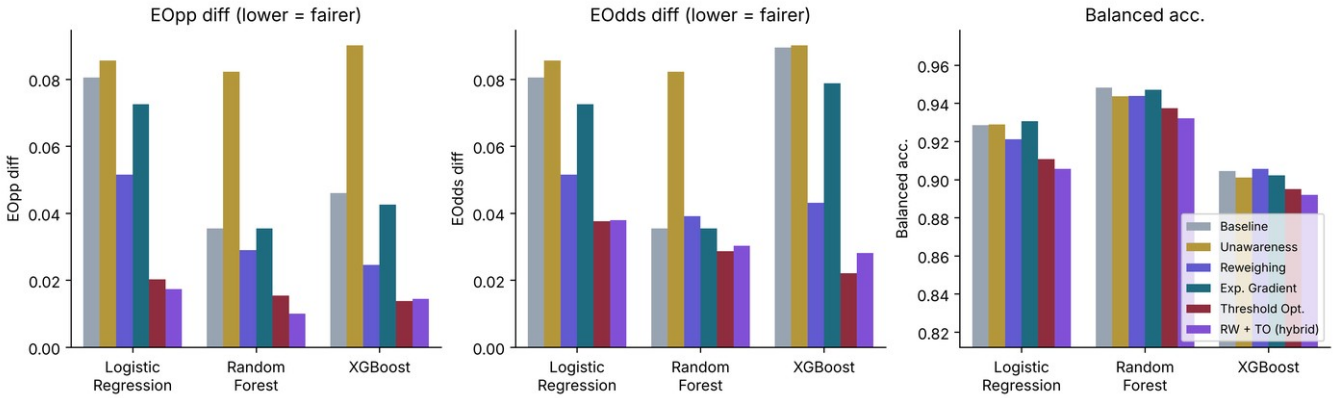
4. Bias mitigation

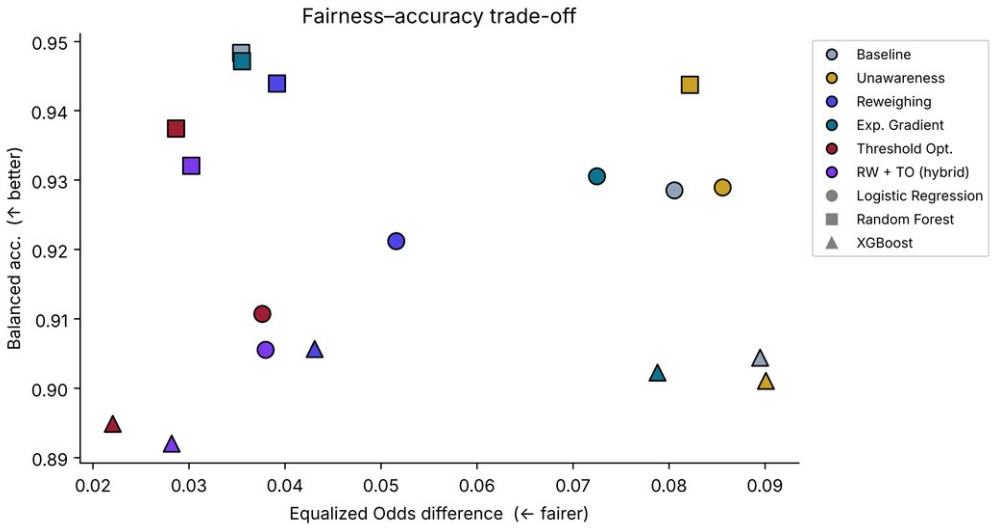
a) Performance (mean ± SD over CV repeats)

Model	Method	Accuracy	Balanced acc.	Recall	F1
Logistic Regression	Baseline	0.912 ± 0.001	0.929 ± 0.001	0.857 ± 0.002	0.923 ± 0.001
Logistic Regression	Unawareness	0.912 ± 0.003	0.929 ± 0.002	0.858 ± 0.005	0.923 ± 0.003
Logistic Regression	Reweighing	0.909 ± 0.005	0.921 ± 0.006	0.872 ± 0.005	0.923 ± 0.005
Logistic Regression	Exp. Gradient	0.914 ± 0.002	0.931 ± 0.002	0.861 ± 0.004	0.925 ± 0.002
Logistic Regression	Threshold Opt.	0.892 ± 0.009	0.911 ± 0.008	0.832 ± 0.014	0.905 ± 0.009
Logistic Regression	RW + TO (hybrid)	0.888 ± 0.015	0.906 ± 0.015	0.834 ± 0.019	0.902 ± 0.014
Random Forest	Baseline	0.940 ± 0.004	0.948 ± 0.004	0.914 ± 0.005	0.950 ± 0.003
Random Forest	Unawareness	0.931 ± 0.002	0.944 ± 0.002	0.890 ± 0.002	0.941 ± 0.002
Random Forest	Reweighing	0.938 ± 0.005	0.944 ± 0.006	0.917 ± 0.003	0.948 ± 0.004
Random Forest	Exp. Gradient	0.940 ± 0.002	0.947 ± 0.003	0.916 ± 0.004	0.949 ± 0.002
Random Forest	Threshold Opt.	0.929 ± 0.005	0.937 ± 0.005	0.902 ± 0.009	0.940 ± 0.004
Random Forest	RW + TO (hybrid)	0.925 ± 0.007	0.932 ± 0.005	0.902 ± 0.014	0.937 ± 0.007
XGBoost	Baseline	0.902 ± 0.006	0.904 ± 0.006	0.893 ± 0.006	0.918 ± 0.005
XGBoost	Unawareness	0.897 ± 0.007	0.901 ± 0.008	0.883 ± 0.009	0.914 ± 0.006
XGBoost	Reweighing	0.903 ± 0.009	0.906 ± 0.008	0.896 ± 0.011	0.920 ± 0.008
XGBoost	Exp. Gradient	0.900 ± 0.009	0.902 ± 0.010	0.892 ± 0.007	0.917 ± 0.007
XGBoost	Threshold Opt.	0.887 ± 0.007	0.895 ± 0.010	0.864 ± 0.011	0.905 ± 0.006
XGBoost	RW + TO (hybrid)	0.884 ± 0.010	0.892 ± 0.012	0.858 ± 0.007	0.901 ± 0.008

b) Fairness gaps (lower = fairer; DI ratio: 1 = fair)

Model	Method	EOpp diff	EOdds diff	PP diff	DP diff	DI ratio	Acc. gap
Logistic Regression	Baseline	0.081 ± 0.004	0.081 ± 0.004	0.000 ± 0.000	0.279 ± 0.002	0.602 ± 0.003	0.015 ± 0.002
Logistic Regression	Unawareness	0.086 ± 0.007	0.086 ± 0.007	0.000 ± 0.000	0.283 ± 0.004	0.598 ± 0.005	0.018 ± 0.004
Logistic Regression	Reweighing	0.052 ± 0.010	0.052 ± 0.010	0.040 ± 0.010	0.246 ± 0.005	0.649 ± 0.007	0.018 ± 0.009
Logistic Regression	Exp. Gradient	0.072 ± 0.007	0.072 ± 0.007	0.000 ± 0.000	0.275 ± 0.004	0.608 ± 0.005	0.011 ± 0.004
Logistic Regression	Threshold Opt.	0.020 ± 0.023	0.038 ± 0.024	0.008 ± 0.008	0.220 ± 0.024	0.664 ± 0.025	0.053 ± 0.025
Logistic Regression	RW + TO (hybrid)	0.017 ± 0.012	0.038 ± 0.014	0.021 ± 0.016	0.222 ± 0.021	0.665 ± 0.023	0.041 ± 0.026
Random Forest	Baseline	0.035 ± 0.009	0.035 ± 0.009	0.023 ± 0.005	0.254 ± 0.003	0.651 ± 0.005	0.011 ± 0.006
Random Forest	Unawareness	0.082 ± 0.004	0.082 ± 0.004	0.004 ± 0.005	0.288 ± 0.003	0.605 ± 0.005	0.026 ± 0.003
Random Forest	Reweighing	0.029 ± 0.006	0.039 ± 0.012	0.038 ± 0.013	0.243 ± 0.006	0.666 ± 0.008	0.015 ± 0.008
Random Forest	Exp. Gradient	0.036 ± 0.004	0.036 ± 0.004	0.024 ± 0.007	0.254 ± 0.006	0.652 ± 0.006	0.012 ± 0.004
Random Forest	Threshold Opt.	0.015 ± 0.012	0.029 ± 0.009	0.028 ± 0.009	0.235 ± 0.014	0.670 ± 0.013	0.009 ± 0.009
Random Forest	RW + TO (hybrid)	0.010 ± 0.009	0.030 ± 0.014	0.031 ± 0.015	0.233 ± 0.006	0.675 ± 0.006	0.011 ± 0.009
XGBoost	Baseline	0.046 ± 0.012	0.089 ± 0.048	0.035 ± 0.028	0.264 ± 0.025	0.647 ± 0.027	0.017 ± 0.012
XGBoost	Unawareness	0.090 ± 0.022	0.090 ± 0.022	0.049 ± 0.013	0.277 ± 0.019	0.629 ± 0.023	0.043 ± 0.009
XGBoost	Reweighing	0.025 ± 0.012	0.043 ± 0.030	0.070 ± 0.021	0.222 ± 0.014	0.693 ± 0.015	0.020 ± 0.014
XGBoost	Exp. Gradient	0.043 ± 0.012	0.079 ± 0.018	0.038 ± 0.036	0.255 ± 0.021	0.656 ± 0.023	0.017 ± 0.016
XGBoost	Threshold Opt.	0.014 ± 0.009	0.022 ± 0.009	0.051 ± 0.024	0.208 ± 0.007	0.699 ± 0.009	0.023 ± 0.018
XGBoost	RW + TO (hybrid)	0.014 ± 0.009	0.028 ± 0.014	0.051 ± 0.018	0.205 ± 0.011	0.701 ± 0.014	0.024 ± 0.010





c) Statistical significance (paired bootstrap vs. baseline, 1000 resamples; Δ = method – baseline)

Model	Method	Δ EOdds gap	95% CI (EOdds)	p (EOdds)	Δ Balanced acc.	95% CI (Bal. acc.)	p (Bal. acc.)
Logistic Regression	Unawareness	0.007	[0.000, 0.026]	0.788	-0.002	[-0.006, 0.000]	0.766
Logistic Regression	Reweighing	-0.008	[-0.038, 0.045]	0.632	-0.013	[-0.030, 0.002]	0.110
Logistic Regression	Exp. Gradient	-0.015	[-0.040, 0.007]	0.366	0.004	[0.000, 0.010]	0.304
Logistic Regression	Threshold Opt.	-0.044	[-0.102, 0.058]	0.230	-0.012	[-0.026, 0.002]	0.102
Logistic Regression	RW + TO (hybrid)	-0.033	[-0.092, 0.038]	0.300	-0.010	[-0.028, 0.009]	0.296
Random Forest	Unawareness	0.031	[-0.029, 0.084]	0.260	-0.000	[-0.016, 0.018]	0.942
Random Forest	Reweighing	-0.004	[-0.023, 0.007]	1	0.002	[0.000, 0.006]	0.772
Random Forest	Exp. Gradient	0.000	[0.000, 0.000]	1	0.000	[0.000, 0.000]	1
Random Forest	Threshold Opt.	-0.002	[-0.032, 0.037]	0.918	-0.013	[-0.026, -0.002]	0.016
Random Forest	RW + TO (hybrid)	0.005	[-0.048, 0.088]	1	-0.015	[-0.029, -0.004]	0.002
XGBoost	Unawareness	0.013	[-0.069, 0.095]	0.702	-0.004	[-0.033, 0.025]	0.772
XGBoost	Reweighing	0.015	[-0.035, 0.074]	0.772	-0.004	[-0.017, 0.006]	0.448
XGBoost	Exp. Gradient	0.009	[-0.009, 0.040]	0.966	-0.007	[-0.018, 0.000]	0.290
XGBoost	Threshold Opt.	-0.014	[-0.092, 0.075]	0.710	-0.012	[-0.035, 0.010]	0.312
XGBoost	RW + TO (hybrid)	-0.010	[-0.091, 0.071]	0.778	-0.016	[-0.036, 0.003]	0.130

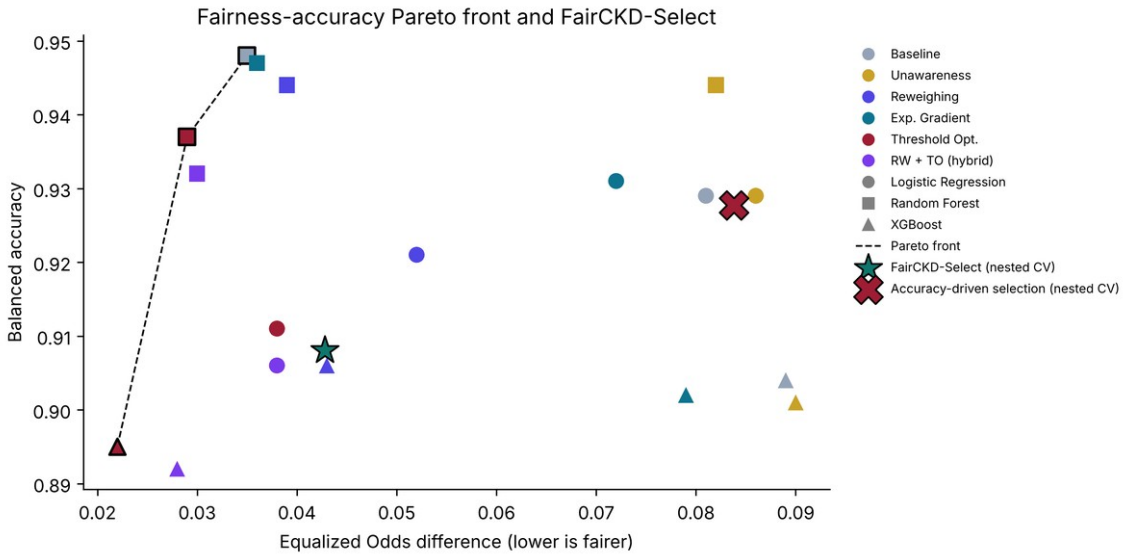
4b. FairCKD-Select (proposed method)

FairCKD-Select chooses, inside each training fold, the classifier and mitigation method with the smallest Equalized Odds gap among those whose balanced accuracy is within 3 points of the best unmitigated model (nested cross-validation). It is compared with conventional accuracy-driven selection.

Strategy	Accuracy	Balanced acc.	Recall	EOpp diff	EOdds diff	DP diff
FairCKD-Select	0.893 ± 0.011	0.908 ± 0.001	0.843 ± 0.041	0.043 ± 0.011	0.043 ± 0.011	0.212 ± 0.042
Accuracy-driven selection	0.910 ± 0.000	0.928 ± 0.000	0.855 ± 0.000	0.084 ± 0.000	0.084 ± 0.000	0.281 ± 0.000

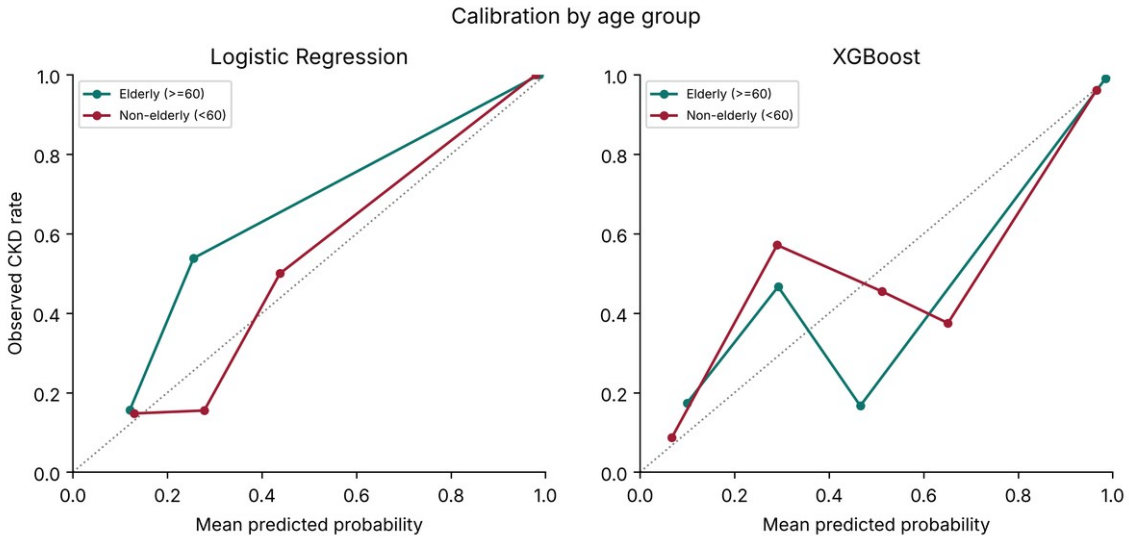
Pipelines chosen by FairCKD-Select across outer folds

Model	Method	Times selected
Logistic Regression	RW + TO (hybrid)	3
Logistic Regression	Reweighing	1
Logistic Regression	Threshold Opt.	5
XGBoost	Reweighing	1

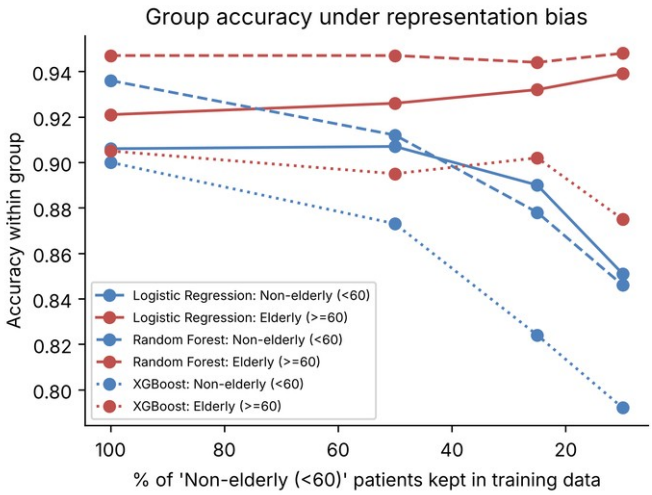


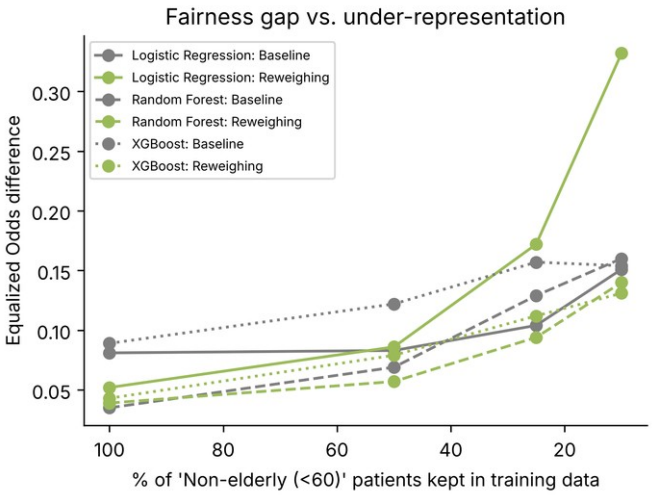
4c. Calibration by group

Group	Patients	Observed CKD (%)	Mean predicted (%)	Calibration-in-the-large (pp)	ECE	Brier	Model
Elderly (>=60)	151	78.146	73.704	-4.441	0.044	0.057	Logistic Regression
Non-elderly (<60)	240	51.667	53.396	1.73	0.056	0.086	Logistic Regression
Elderly (>=60)	151	78.146	75.81	-2.336	0.047	0.067	XGBoost
Non-elderly (<60)	240	51.667	50.56	-1.107	0.038	0.084	XGBoost

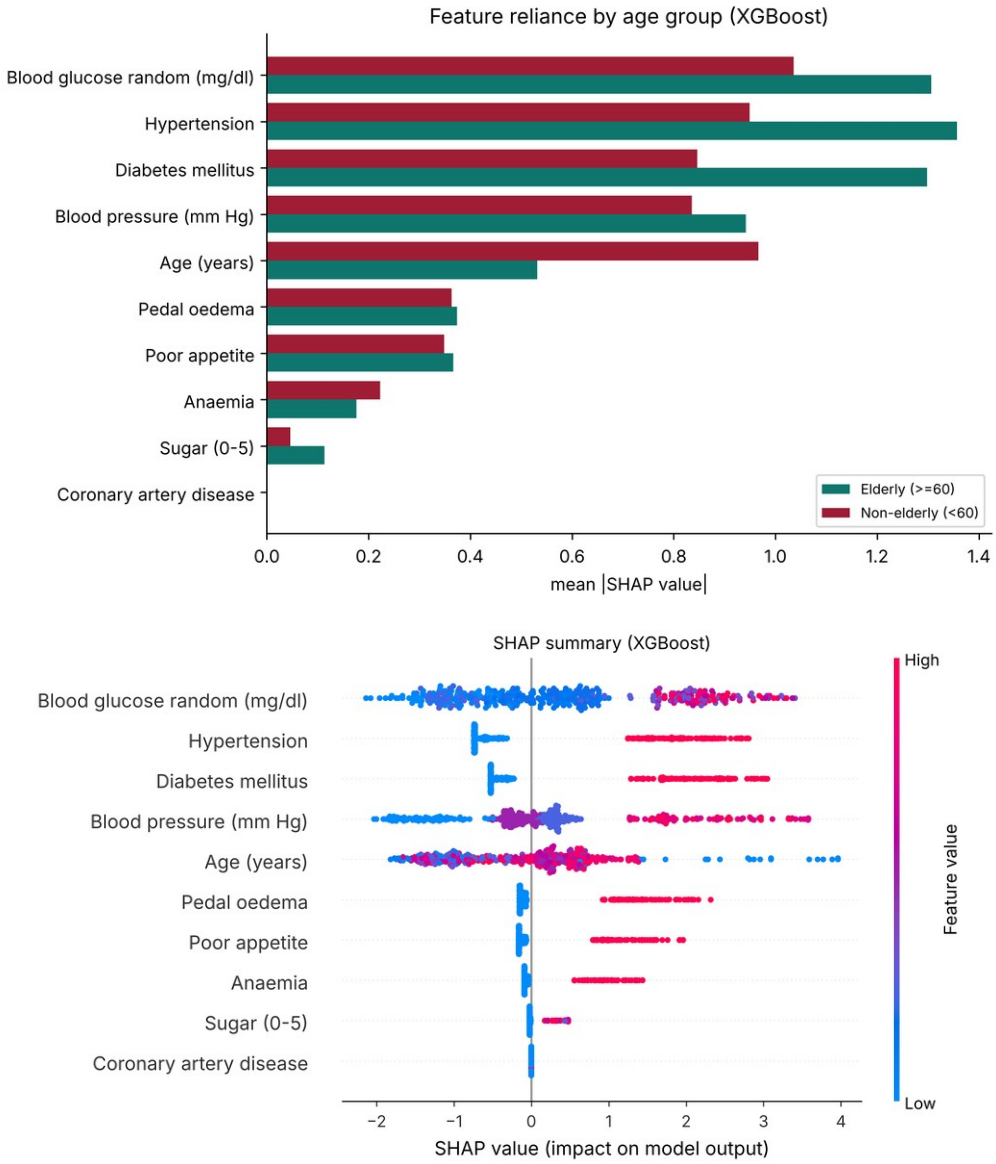


5. Representation-bias experiment





6. Explainability (SHAP)



Protocol: repeated stratified 5-fold cross-validation (stratified on label × group); preprocessing and mitigation fitted on training folds only; mean ± SD over repeats. This system is a research prototype and not a medical device.